25900688 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZGYA00 - BIOLOGY OF GENOMES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Compared with animals and plants, many fungal nuclear genomes are: [U] / [CO1]

   a). Uniformly >1 Gb and intron-rich
   b). Typically compact and gene-dense, with fewer introns on average
   c). Organized in polycistronic operons like bacteria
   d). Composed mostly of chloroplast-derived DNA

2. The Arabidopsis thaliana genome is best described as: [K] / [CO1]

   a). Polyploid with many chromosome sets
   b). Diploid with 5 chromosomes
   c). Triploid with 2 chromosome sets
   d). Haploid

3. Which organelle is thought to have originated from an endosymbiotic event with a photosynthetic cyanobacterium? [K] / [CO1]

   a). Nucleus b).Mitochondrion c). Chloroplast d). Golgi apparatus

4. Which process allows plasmids to move directly from one bacterial cell to another? [U] / [CO2]

a). Transduction
b). Conjugation
c). Transformation
d). Binary fission

5. Which of the following statements about bacterial chromosome organization is correct? [A] / [CO2]

a). The bacterial chromosome is linear like eukaryotes
b). Supercoiling and NAP interactions condense the chromosome
c). Histone proteins are essential for chromosome packaging in bacteria
d). Nucleoid is enclosed by a double membrane

6. Why was sequencing the SARS-CoV-2 genome important for vaccine development? [E] / [CO3]

a). It helped identify the spike protein as a target
b). It allowed direct production of antibodies
c). It replaced the need for clinical trials
d). It stopped the virus from mutating

7. What does the "+ssRNA" genome of Tobacco Mosaic Virus mean? [E] / [CO3]

a). It must be converted into DNA before it can be used for protein synthesis(translation)
b). It can directly function as mRNA for protein synthesis
c). It cannot be translated
d). It forms double-stranded RNA first

8. Why is Bayesian inference increasingly used in phylogenetics? [An] / [CO4]

a). It provides fixed answers
b). It uses statistical probability to estimate tree confidence
c). It ignores evolutionary models
d). It excludes molecular data

9. How is the Geological Time Scale constructed? [An] / [CO4]

a). By analyzing atmospheric gases
b). By radiometric dating and fossil records
c). By studying ocean currents
d). By satellite imaging

10. The total chromosome number in chimpanzees is: [K] / [CO5]

a). 42 b). 44 c). 46 d). 48

[1x10 = 10]

Part B

Short Answer Questions

Answer any **14** questions

Each question carries **2** marks

11. What percentage of the human genome is estimated to be protein-coding? [K] / [CO1]

12. How do mitochondrial ribosomes link mitochondria to bacterial ancestors? [U] / [CO1]

13. Like mitochondria, chloroplasts are also believed to have an endosymbiotic origin. Mention its probable ancestor? [K] / [CO1]

14. Explain the difference between tandem repeats and interspersed repeats. [U] / [CO1]

15. Explain the concept of integrons and analyze their role in the development and spread of antibiotic resistance in bacteria. [U] / [CO2]

16. How does having minimal non-coding DNA help prokaryotes adapt to their environment? [A] / [CO2]

17. Describe the structure and composition of the bacterial chromosome found in the nucleoid. [U] / [CO2]

18. Mention two general features of prokaryotic genomes. [K] / [CO2]

19. Compare single-stranded and double-stranded viral genomes and evaluate their stability inside the host. [E] / [CO3]

20. How does evaluating variants of SARS-CoV-2 help in deciding monoclonal antibody use? [E] / [CO3]

21. What makes the SARS-CoV-2 genome suitable for rapid mutation and evolution? [E] / [CO3]

22. Which enzyme is used by SARS-CoV-2 to copy its RNA genome? [K] / [CO3]

23. What is a haplotype block? [U] / [CO4]

24. State an application of molecular clocks. [A] / [CO4]

25. Define a phylogram. [K] / [CO4]

26. Explain the difference between rooted and unrooted phylogenetic trees. [U] / [CO4]

[2x14 = 28]

Part C

Short Essay Questions

Answer any **8** questions

Each question carries **4** marks

27. Discuss how histone proteins contribute to the packaging of DNA in the human genome [U] / [CO1]

28. Briefly list at least two significant differences between the structure or organization of the mitochondrial genome and the nuclear genome of eukaryotic cells [U] / [CO1]

29. Briefly explain the concept of "jumping genes" in maize and name two distinct types of transposable elements found in this plant. [U] / [CO1]

30. Using the lac operon as an example, outline its structure and explain how it regulates gene expression in response to the presence or absence of lactose. [A] / [CO2]

31. Explain the structure of *E. coli* genome with a simple diagram. [U] / [CO2]

32. Analyze why SV40 is widely used as a model organism in molecular biology and cancer research. [An] / [CO3]

33. What is the role of the membrane (M) protein in the virus? [U] / [CO3]

34. Discuss the role of cladistics in constructing cladograms. [U] / [CO4]

35. Explain how comparative genomics can reveal evolutionary divergence among species. [An] / [CO4]

36. Compare and contrast cladograms, phylograms, and chronograms. [An] / [CO4]

[4x8 = 32]